Test Documentation — Vehicle Washing Management System

Cover Page

Table 1

Document Identification Data

Field	Value
Project	Vehicle Washing Management System (Vehicle Washing / LavaRápido)
Deliverable	Test documentation (section 2.5.4)
Program	Software Analysis and Development Technologist (ADSO)
Group (ficha)	3145556
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Instructor	Karol Daniela Correa
Training center	Industria
Date	November 4, 2026

This document gathers the test plan, the inventory of automated tests in the repositories, their actual results when run on October 19, 2026 (including the end-to-end, performance, and responsive design tests against the complete system) and again on October 21, and the 42 functional test cases, run on October 19, traced to the SRS v8.4 requirements (`lavarapido-docs/01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS`, the current SRS; *SRS_V4.docx* is the previous version of October 8). On October 24, 2026, after merging 128 fix branches into *develop* and deploying them to the public server, the inventory (Table 4), the results of the latest CI (section 3.3, Table 5d), the traceability matrix (eight requirements move to Implemented), and the gaps (Table 8) were updated again; the same day, notification migration 048, which fixed what the live test found in 047, was added to gap 9. On October 26, 2026 the live tests of payment at booking were added, locally and in production, with the two simulated failures (section 3.4, Table 5e), the new test count of booking, payment, the web, and the mobile app (Table 5f), defect 13 (Table 9), and gaps 9 and 10 (Table 8). On October 27, 2026, against `main` of that day, the counts (Table 5f: 1,337 tests, with payment's 17 integration tests, which now run in the CI against SQL Server), the end-to-end run (section 3.1: 19 of 19), the booking step of CP-08, the new cases CP-43 to CP-46, and gaps 1 to 3 (Table 8) were updated. On November 4, 2026 the payment-service tests were updated with *v1.1.3*, which was merged into `main` on October 27 and passed the CI that day (Tables 4 and 5f and gap 3: 275 in payment and 1,348 in total). On October 23, 2026 the inventory (Table 2), the results of the latest *develop* CI (section 3.3, Table 5d), and the traceability matrix were updated; the matrix now marks as partial the requirements that the review of that day found incomplete. On October 20, 2026, the inventory, case CP-42 and the traceability matrix were adjusted because the role change of an account was reverted in the code that day, and the wording of CP-10 because the customer cancels without writing a reason. On October 21, 2026, two runs of the automated tests and of the 19 end-to-end tests were recorded: the early-morning run (689 tests, all passed) and the evening run (707, all passed), which is the latest. They include the changes of that day: the token of a closed session stops working, operator assignment uses a lock, the receipt has a 2 MB maximum, and, for bookings that ended and nobody closed, the system only cancels a scheduled booking that expired without confirmation and sets to "No-show" a confirmed one that never had an operator; one that had an operator goes to the administrator's "To close" list. Its structure follows the ISO/IEC/IEEE 29119-3 standard for test documentation (International Organization for Standardization [ISO], 2021). No result in this document is assumed.

1. Test Plan

1.1 Scope

The web application, the mobile application, and the seven backend services on their `develop` branch are tested. Offensive security testing is out of scope; load is only checked through the RNF-001 response time measurement.

1.2 Test Types and Tools

Table 2

Test Types, Actual Tool, and Where Each Applies

Type	What it verifies	Tool	Where
Domain unit tests	Rules of entities and value objects (states, validations)	JUnit 5 (`spring-boot-starter-test`); xUnit 2	`src/test/.../domain` in Java; `tests/PaymentService.Domain.UnitTests`
Application unit tests	Use cases with in-memory repositories (test doubles)	JUnit 5 with fakes (`Fakes.java`, `InMemory*Repository`); xUnit with `InMemoryPorts.cs`	`application/usecase` and `PaymentService.Application.UnitTests`
Web integration (controller + security)	Routes, HTTP status codes, JWT, and roles with MockMvc	`spring-boot-starter-webmvc-test`, `spring-security-test`	`*ApiWebTest.java`, `ApiSecurityWebTest.java`, `AdminNotificationControllerTest.java`
Gateway integration	Each public route reaches its service and the token of a closed session does not get through	Spring Boot Test; for the session, a security-service simulated with the JDK HTTP server	`GatewayRoutesTest.java`, `SessionCheckFilterTest.java`
Persistence integration	EF Core mapping against a real SQL Server	xUnit + `DatabaseFixture`	`PaymentService.Infrastructure.IntegrationTests` (requires a database)
Web UI unit tests	Angular components, guards, and services	Vitest 4 + jsdom (`@angular/build:unit-test`)	`Web/src/**/*.spec.ts`
Functional / acceptance	End-to-end flows per role against the deployed system	Manual, with Postman or Swagger UI for the API and a browser / Expo Go for the UI	Cases CP-01 to CP-42 (section 4)
Cross-service (end-to-end) and RNF	Booking with operator choice and automatic assignment, complete flow of the three roles, inspection report, access rules per role; 95th percentile response time (RNF-001); web screens at 360 px (RNF-008)	Node 24 (`node:test` and `fetch`) and Playwright Core with headless Edge	`lavarapido-infra/e2e`

Note. Tools documented in JUnit (n.d.), xUnit.net (n.d.), and Vitest (n.d.).

1.3 Environment

Table 3

Execution Environment for the Tests (October 19, 2026)

Item	Version
Operating system	Windows 11 Home Single Language 10.0.26300
Node.js / npm	24.14.0 / 11.9.0
JDK	Java 25.0.4 (the projects compile for Java 21) with each service's Maven Wrapper
.NET SDK	9.0.306; the project targets `net8.0` and the tests were run with `DOTNET_ROLL_FORWARD=Major`
Database and messaging	SQL Server 2022, RabbitMQ 4.3, and Mailpit on Docker 28.4 (Docker Desktop). The unit tests do not need them; the 7 payment persistence tests used an isolated database, `LavaRapidoPaymentIT`, created with the same changelogs
Services	The six services and the gateway running on the machine (`./mvnw spring-boot:run` and `dotnet run`) against that infrastructure; email went to Mailpit so no real person was written to. The web with `ng serve` and the mobile app with `npx expo start --web`

1.4 Criteria

- **Passed:** the actual result matches the expected one.
- **Failed:** the result differs; the error is recorded.
- **Blocked:** could not run because a dependency was missing (for example, the database).
- **Exit:** the release is accepted when the automated suites pass and the high-priority functional cases are passed.

2. Automated Test Inventory

As of October 24, 2026 (*develop* branch), the repositories have 1,036 test methods: 759 in the backend, 204 in the web (212 cases when run, because some repeat with several data sets), and 73 in the mobile app (on October 23 there were 735: 576, 146, and 13; on October 21, 602). The mobile app has tests with *jest-expo* since October 23; it is also checked with the TypeScript build and the screen walk-through (section 3.2).

Table 4

Tests per Artifact (Count of `@Test`/`@ParameterizedTest`, `[Fact]`/`[Theory]`/`[IntegrationFact]`, and `it()`)

Artifact	Files	Methods	What they cover
api-gateway	6	38	Routing of each `/api/v1/**` prefix to its service, and that `/internal/**` is not exposed by the gateway (`GatewayRoutesTest`); session check with any scheme (*bearer*, *Bearer*, *BEARER*, *Basic*), the answer is kept per token and logout drops it in all its variants (`SessionCheckFilterTest`); a limit of 60 requests per minute on `/auth` with the real IP (`AuthRateLimitFilterTest`, `ClientIpResolverTest`, `GatewayAuthRateLimitTest`) and route timeouts (`GatewayRouteTimeoutTest`)
security-service	19	168	Registration with mandatory consent (`CONSENT_REQUIRED`), sign-in with lockout after failed attempts, a cap on the attempts for the recovery code and the current password (`TOO_MANY_ATTEMPTS`), passwords of up to 72 bytes, the client IP behind the proxy (`ClientIpBehindGatewayTest`), password recovery, profile and address, user administration (including the rule that an administrator account cannot be deactivated), roles and permissions, preferences, value objects, password policy, JWT issuing, open and closed sessions (the token of a closed session answers 401), and HTTP security by role (`ApiSecurityWebTest`)
customer-service	8	61	Customer account, license plates (strict when creating and tolerated when reading stored ones), vehicle CRUD, vehicle with active bookings (`VEHICLE_HAS_ACTIVE_BOOKINGS`), customer-only routes, `security.user_registered` listener and web API
booking-service	14	95	Customer rescheduling without changing services (`BOOKING_SERVICES_LOCKED`), price greater than 0, optimistic locking, frozen rate per vehicle type, schedule changes that clash with bookings (`ScheduleServiceTest`), forwarded customer errors, active bookings count per vehicle, booking life cycle, cancellation of scheduled bookings that expired without confirmation, "No-show" only for an expired confirmed booking that operations requests through the internal route, availability and the reason a time is not available, bays (`ServiceBayTest`), catalog, business data (`EstablishmentTest`), preferred operator in the event and web API
operations-service	20	109	Starting and finishing without leaving the operator blocked if booking fails (OPS-01), row locks, start window up to 4 hours after the end (`SERVICE_TOO_LATE`), notes for the operator, listener retries and dead-letter queue, sharing the inspection without a broker (503), database errors, manual and automatic assignment (with the assignment lock), a single rule for bookings without an operator, bookings that ended and nobody closed and the "To close" list (`FollowUpPolicyTest`, `FollowUpServiceTest`), free operators, start window, execution, rating, absences, operator metrics, team summary, operator profile, inspection report and web API
notification-service	18	127	Pending deliveries with retries (`pending_delivery`), reminders cancelled by IN_PROGRESS, COMPLETED, or NO_SHOW, released-operator notice, administrator lookup with retries, bindings and dead-letter queue, notification use cases, message factory per event, notices in the account's language, reminders, alerts for bookings without an operator, unregistered service notice, `null` fields in the event (`DomainEventEnvelopeTest`), new version notice, device tokens and admin controller
payment-service (.NET)	14	161	The customer's amount and coupon (`GET /loyalty/redemptions/{bookingId}`), `AMOUNT_CHANGED` when approving, a percentage coupon with the current total, a single open payment and a unique reference, the administrator's paged queue, cancelled-booking notice, inactive cash for the administrator, API tests by role (`PaymentService.Api.UnitTests`), payment and receipt (2 MB maximum), reused reference, history with filters, promotions, loyalty points, payment policy, command and query services, and the persistence tests against SQL Server (11 as of October 24; 17 integration tests since October 27, 2026, which run in the CI, and 25 with *v1.1.3*, which adds those for the connection without `DB_PASSWORD` and for the RabbitMQ credentials)
Web (Angular)	66	204	Real guard and interceptor with `app.routes` (role × area, expired token, 401 and 403), payment with the server's amount and the coupon that survives a reload, automatic refresh (`LiveRefresh`), the 4 languages with the same keys and translated error codes, field errors, the operator's notes and start window, `BOOKING_NO_SHOW`, the sign-up, payment, and dashboard ViewModels; creation of components, guards and services (`should create` smoke tests), API error messages, payment account selection, session across tabs, bookings without an operator and filters, "To close" list, accounts and fixed roles, reports, and loyalty bar
Mobile	25	73	With *jest-expo*: validators (`authValidators.test.ts`), the customer's payment with the server's amount (`useClientPayment.test.tsx`), refresh on return and on each push (`useRefreshOnFocus.test.tsx`), error codes (`apiError.test.ts`), the operator contract (`opsContract.test.ts`), the paged payment queue (`paymentQueue.test.tsx`), fixed discount (`promotionDiscount.test.ts`), production URL (`config.test.ts`), recovery with "Request another code" (`useForgotPasswordFlow.test.tsx`), session and push unregistration (`AuthServiceContext.test.tsx`), accessibility of icon buttons, and pagination

Note. Support files (`Fakes.java`, `InMemory*.java`, `InMemoryPorts.cs`, `DatabaseFixture.cs`) are not counted as tests. The 11 payment persistence tests skip themselves when the `PAYMENT_INTEGRATION_DB` variable is not set (`DatabaseFixture.cs:16-21`). Since October 27, 2026 the CI runs the integration project in the "Pruebas de integración" job, against SQL Server 2022 with the Liquibase migrations, without

skipping any: 17 tests with *v1.1.2* and 25 with *v1.1.3*, of which 12 use the database and 13 do not; on a machine without a database, those 12 are skipped.

Backend tests are domain and application tests with in-memory doubles, plus controller tests with MockMvc. Many web tests only check that each component is created; `api-error.spec.ts`, `payment.spec.ts` and, since October 21, 2026, `auth.spec.ts`, `reservations-store.spec.ts`, `reservations.spec.ts`, `management.spec.ts`, `reports.spec.ts`, and `dashboard.spec.ts` test real rules.

3. Execution Results

On the evening of October 21, 2026, with the *develop* branch of the ten repositories updated that day, the team ran the eight automated suites again: 707 tests, all passed and none skipped, including the 7 payment persistence tests against the isolated `LavaRapidoPaymentIT` database. It is the final run of the day and includes the change to bookings that ended and nobody closed, made after the team review, and the fix for `null` fields in notification-service events. Table 5 shows each suite and what changed from the early-morning run.

Table 5

Actual Results per Artifact (October 21, 2026, Evening Run)

Artifact	Command	Run	Passed	Failed	Skipped	Compared with the early-morning run of October 21 (689)
Web (Angular)	`npx ng test --watch=false`	62	62	0	0	+2: "To close" list in the bookings store (`reservations-store.spec.ts`)
api-gateway	`./mvnw test`	40	40	0	0	No change
security-service	`./mvnw test`	155	155	0	0	No change
customer-service	`./mvnw test`	69	69	0	0	No change
booking-service	`./mvnw test`	50	50	0	0	+3: only a scheduled booking expires by itself, and a confirmed one becomes No-show only when operations asks for it through the internal route and it has already expired (`BookingTest`, `BookingServiceTest`, `BookingApiWebTest`)
operations-service	`./mvnw test`	67	67	0	0	+11: what to do with a booking that ended and nobody closed (`FollowUpPolicyTest`), closing the ones that never had an operator, a single notice, the "To close" list, and closing as "Attended" without made-up times (`FollowUpServiceTest`, `ServiceExecutionTest`, `OperationsApiWebTest`)
notification-service	`./mvnw test`	114	114	0	0	+2: notice to the operator and the administrator about an unregistered service (`StaffNotificationsTest`) and a `null` field in the event that no longer goes to the dead-letter queue (`DomainEventEnvelopeTest`)
payment-service: domain	`dotnet test PaymentService.sln`	115	115	0	0	No change
payment-service: application	(same command)	28	28	0	0	No change
payment-service: integration	(same command, with `PAYMENT_INTEGRATION_DB`)	7	7	0	0	No change
Total		**707**	**707**	**0**	**0**	**+18**

Note. Sources: records of the team runs of October 21, 2026 (`vehicle-w-docs/11-quality/testing-strategy.md`), Vitest and `dotnet test` output, and the `./mvnw test` summary (`Tests run:`) with the surefire reports. More tests run than there are methods in Table 4 because parameterized tests (`@ParameterizedTest`, `[Theory]`) run once per data item. In customer-service the `tests` attribute of the surefire XML says 68 because of a nested test class; Maven runs and counts 69, which is the right number.

The early-morning run of October 21, 2026 had 689 tests, all passed: web 60, api-gateway 40, security-service 155, customer-service 69, booking-service 47, operations-service 56, notification-service 112, and payment-service 150 (115 domain, 28 application, and 7 persistence). Compared with October 19 it added 34: session across tabs, bookings without an operator and filters, accounts and fixed roles, reports, and loyalty bar in the web (+17); session check in the gateway (+6); sessions in security-service (+4, after removing 2 for the reverted role change);

expired bookings in booking-service (+4); assignment lock in operations-service (+1); and the 2 MB receipt maximum in payment-service (+2). That same early morning, the suites that had changed were repeated separately, with the same result.

The run of October 19, 2026 had 655 tests, all passed: web 43, api-gateway 34, security-service 151, customer-service 69, booking-service 43, operations-service 55, notification-service 112, and payment-service 148 (113 domain, 28 application, and 7 persistence). Figures 1 to 3 are from that run.

Besides the automated tests, the "To close" list was checked live on October 21, 2026: a booking without an operator became No-show by itself; one with an operator that was never started and one started but never finished were left to close, with one notice to the operator and another to the administrator that was not repeated; closing them as "Attended" left them completed without made-up times, and closing them again answered 409. It looked the same on the web and in the mobile app (390 px).

Figure 1

Result of the Java Service Tests (October 19, 2026)

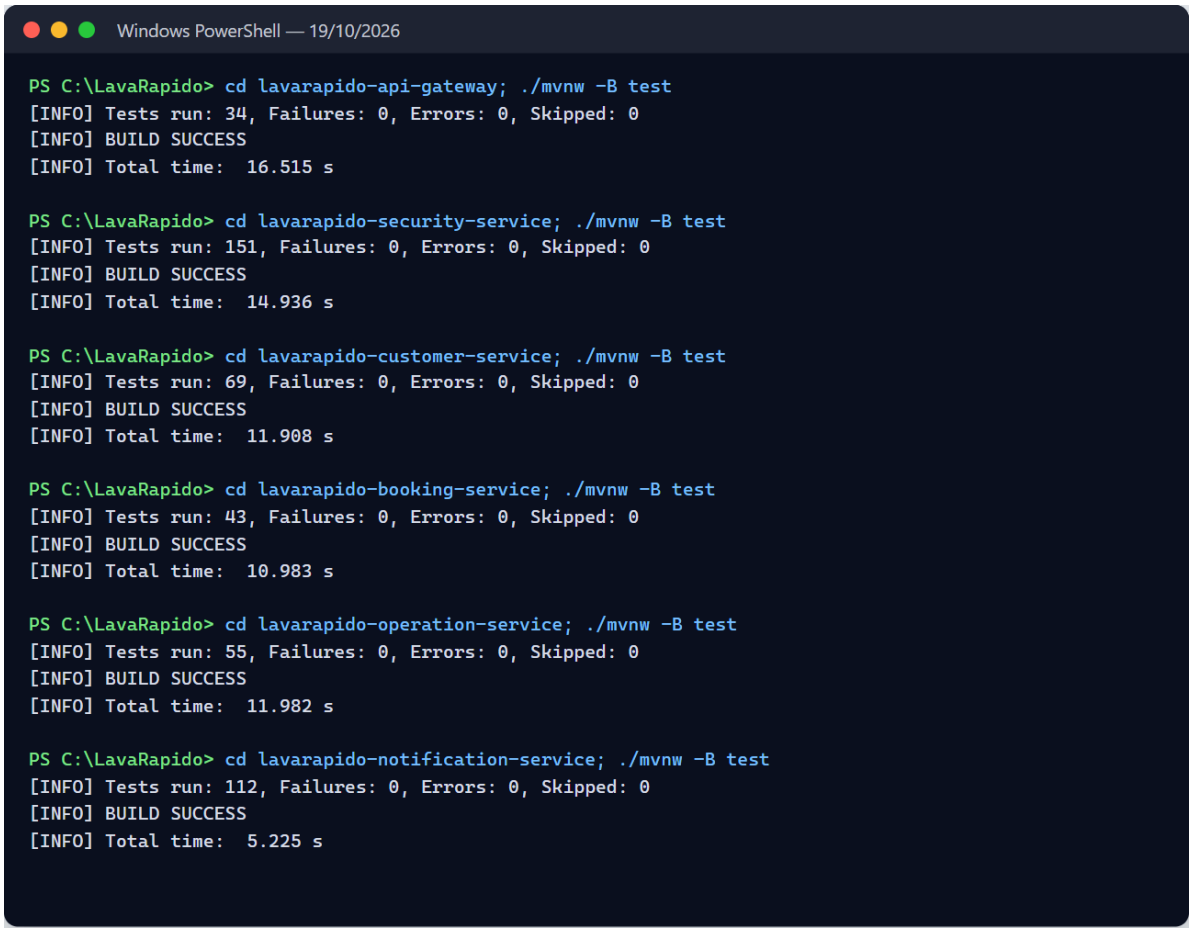


Figure 2

Result of the payment-service Tests, Including the Persistence Ones (October 19, 2026)

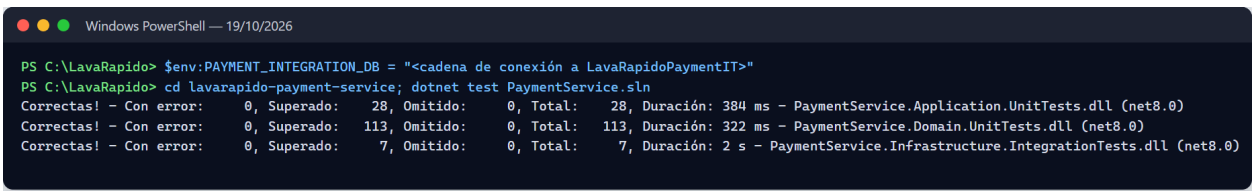
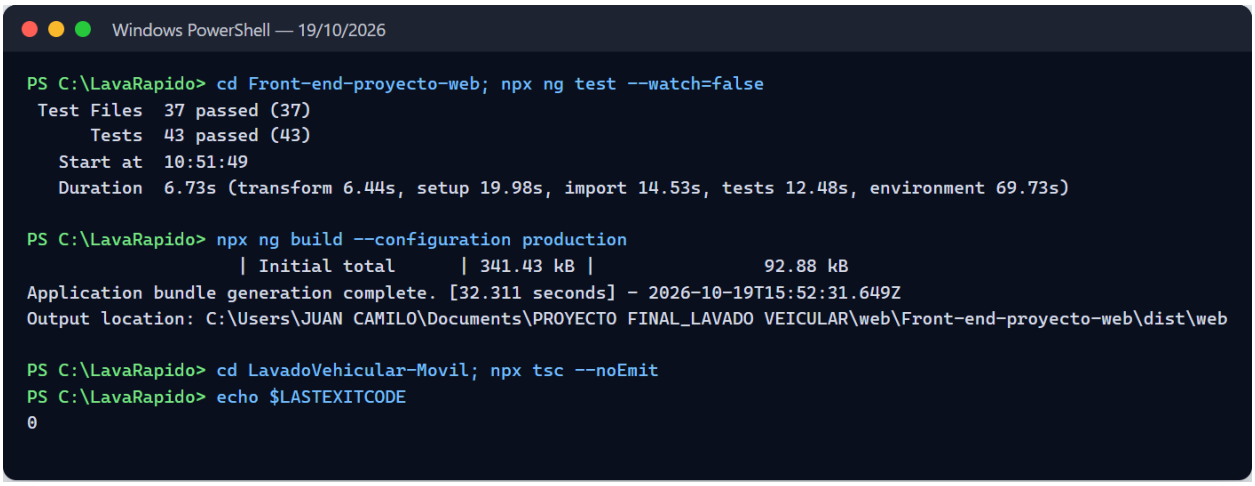


Figure 3

Web Tests and Production Build, and TypeScript Build of the Mobile App (October 19, 2026)



Execution remarks:

- The line `ERROR ... Could not send the password recovery email: SMTP down` in the security-service log is caused on purpose by `EmailResetCodeNotifierTest`, which simulates a down SMTP server. It is not a failure.
- Vitest shows a build warning: `src/test-providers.ts` is not in the `tsconfig`. It does not affect the results.
- The October 19 run used JDK 25 and .NET SDK 9 on projects for Java 21 and .NET 8. GitHub Actions runs the same tests with Temurin 21 and .NET 8 on every change (Deployment Document).

3.1 End-to-End Tests

On October 19, 2026, `node --test "e2e/*.test.js"` was run from `lavarapido-infra`, with the complete system running and `E2E_OPERATOR_ID=2` (the test operator of the local database). The 19 tests passed in 11.2 s. Each test deletes what it creates and leaves the customer's points as they were. On October 21, 2026, the team ran them again, early in the morning and in the evening, with the gateway session check on, and all 19 passed both times; the files in Table 5a have not changed since October 18. On October 27, 2026 they were run again against the local system with the `main` code and passed 19 of 19, after the suite was adjusted to send the payment mode (`paymentMode`) when booking: booking-service requires it since October 26, and without it the tests that book failed with 400 `INVALID_PAYMENT_MODE`.

Table 5a

End-to-end tests (October 19, 2026)

File	Tests	What it walks through	Result
`access-rules.test.js`	7	Permissions by role and internal routes, profile address, payment history filters, tax data only for the administrator, operator profile, time outside opening hours (`reason: OUTSIDE_HOURS`), and the token of an account that does not exist	7 passed
`booking-flow.test.js`	5 (1 + 4 subtests)	Booking choosing an operator, automatic assignment through RabbitMQ, customer notices, day summary, and internal routes outside the gateway	5 passed
`full-flow.test.js`	6 (1 + 5 subtests)	Booking → reported payment → approval with points → service start and finish → rating, checking each notice	6 passed
`inspection-flow.test.js`	1	Inspection report with findings and photos, publishing, customer notice, and public link	1 passed
Total	**19**		**19 passed**

Figure 4

Output of the End-to-end Tests



3.2 Performance and Responsive Design

The same day, the response time of the most used reads (NFR-001) was measured with `node e2e/performance.js`: 30 requests per endpoint through the gateway. All of them were far below the 2 s limit at the 95th percentile.

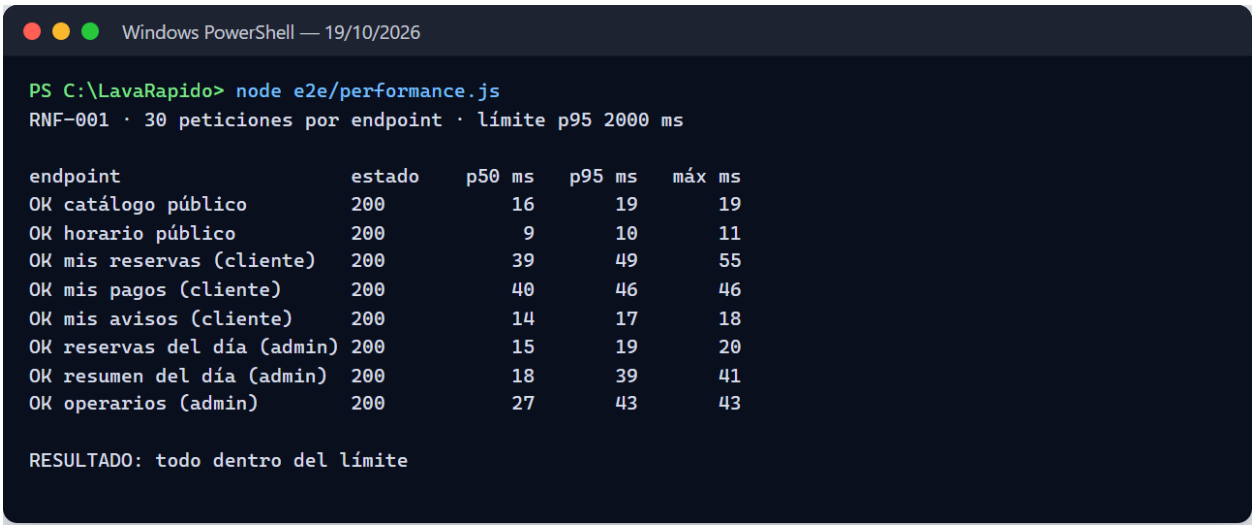
Table 5b

Response time per endpoint (NFR-001, October 19, 2026)

Endpoint	p50	p95	Maximum
Public catalog	16 ms	19 ms	19 ms
Public opening hours	9 ms	10 ms	11 ms
My bookings (customer)	39 ms	49 ms	55 ms
My payments (customer)	40 ms	46 ms	46 ms
My notices (customer)	14 ms	17 ms	18 ms
Day bookings (administrator)	15 ms	19 ms	20 ms
Day summary (administrator)	18 ms	39 ms	41 ms
Operators (administrator)	27 ms	43 ms	43 ms

Figure 5

Response Time Measurement



For responsive design (NFR-008), Playwright (headless Edge) went through the screens of the four web groups (public, customer, operator, and administrator) at 360 and 768 px, with the suite's `width-360.js` script and a check that also looks for elements past the right edge outside a container with its own scrolling.

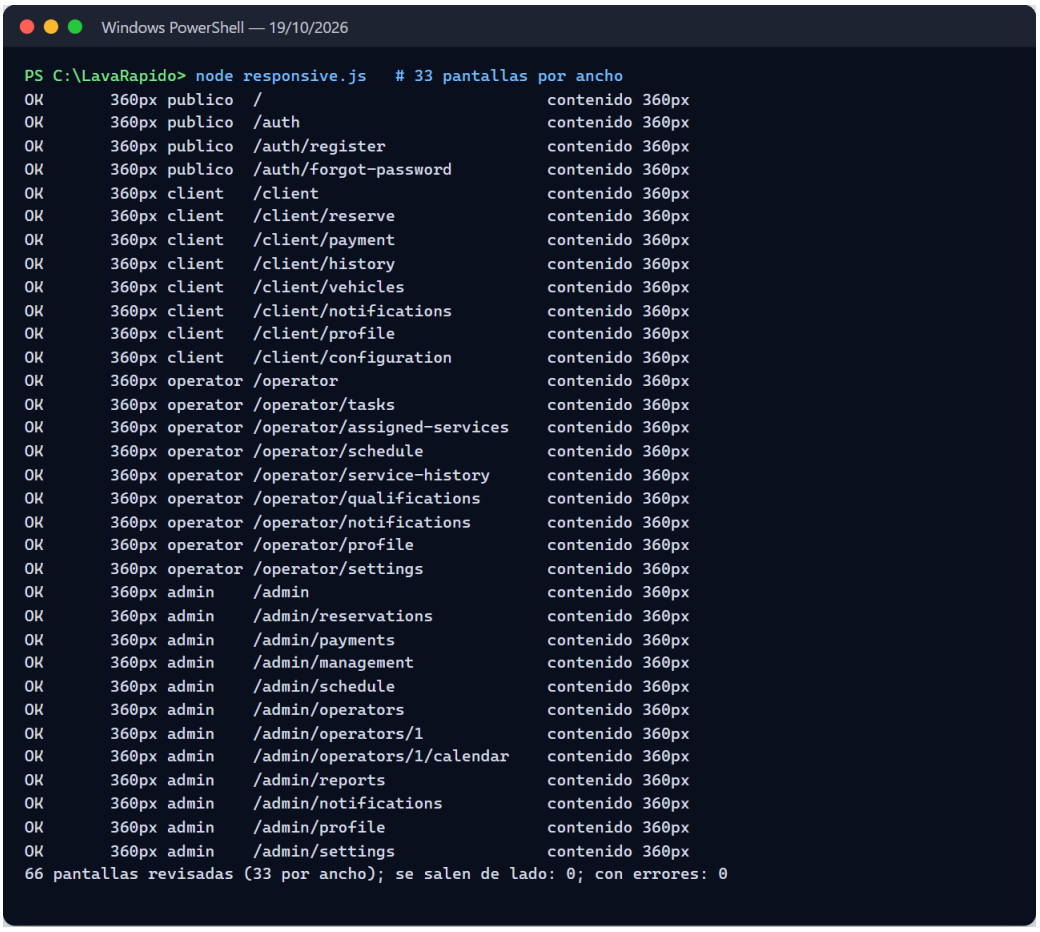
Table 5c

Responsive design check (October 19, 2026)

Test	Scope	Result
Web at 360 px (`e2e/width-360.js`)	12 screens: public, customer, and administrator	None goes beyond 360 px
Web at 360 px	33 screens: 4 public, 8 customer, 9 operator, and 12 administrator	None scrolls sideways or cuts content; no page errors or 5xx responses
Web at 768 px	The same 33 screens	None scrolls sideways or cuts content
Mobile app at 360 px (`npx expo start --web`)	15 screens: the five of the bottom menu of each role (customer, operator, and administrator), with temporary accounts deleted at the end	None scrolls sideways; no page errors
Build	`npx ng build --configuration production` (web) and `npx tsc --noEmit` (app)	No errors

Figure 6

Check of the Web Screens at 360 Px



Note. Taking or picking photos with the phone camera and gallery cannot be automated in this test; it remains to be tested on an Android device.

3.3 Latest CI of October 24, 2026

On October 23, 2026 *v1.0.0* was tagged in 8 repositories (757 tests), and on October 24 the 128 fix branches of the 10 repositories were merged into *develop*. Before pushing each repository, its branches were merged onto *develop* in a separate copy, the full suite was run, and it was checked that what was pushed was identical to what was tested. Table 5d takes the latest GitHub Actions CI of *develop* in each repository on October 24, all green; that same day this code was deployed to the public server. The 19 local end-to-end tests and the 42 functional cases were not run that day; the new remote battery (44 cases, *e2e.yml*) has only been tested against a simulator.

Table 5d

Results of the latest develop CI per artifact (October 24, 2026)

Artifact	Command in the CI	Run	Passed	Failed	Skipped	Compared with October 23 (838)
Web (Angular)	`ng test`	212	212	0	0	+66: server amount, automatic refresh, translations of the 4 languages, field errors, real guard and interceptor, read-only permissions, operator (66 files)
api-gateway	`./mvnw verify`	62	62	0	0	+23: session with any scheme (SEC-01), `/auth` limit and real IP (SEC-04, SEC-11), and timeouts (SEC-16)
security-service	`./mvnw verify`	200	200	0	0	+29: consent (SEC-07), attempt caps (SEC-05, SEC-12), 72 bytes (SEC-11), IP behind the proxy, 403 tests by role (F-04, SEC-09)
customer-service	`./mvnw verify`	82	82	0	0	+11: vehicle with active bookings (BIZ-08), customer-only routes (BIZ-12), stored plates (E-16), and Merari's tests (F-05)
booking-service	`./mvnw verify`	95	95	0	0	+35: BIZ-01, BIZ-04, BIZ-06, BIZ-08, BIZ-09, BIZ-10, BIZ-12, BIZ-13, `reasonCode`, and F-05
operations-service	`./mvnw verify`	110	110	0	0	+38: OPS-01, OPS-03, OPS-04, OPS-08, OPS-11, OPS-12, A-12, A-13, B-01, E-12, and F-05
notification-service	`./mvnw verify`	133	133	0	0	+18: pending deliveries (OPS-05), reminders (A-11), administrators with retries (OPS-06), released operator (E-12)
payment-service: domain	`dotnet test`	134	134	0	0	+15: `AMOUNT_CHANGED`, percentage coupon, cancelled-booking notice
payment-service: application	(same command)	52	52	0	0	+20: customer redemption (BIZ-02), paged queue (E-10), inactive cash (C-04), discount type (D-01)
payment-service: API	(same command)	34	34	0	0	+34: new project with the tests by role (F-04) and the required identifiers (BIZ-13)
payment-service: infrastructure	(same command, without `PAYMENT_INTEGRATION_DB`)	5	5	0	11	+5 without a database; the 11 persistence tests used to be skipped in the CI (since October 27, 2026 they run in the "Pruebas de integración" job, against SQL Server with the Liquibase migrations)
Mobile app	`npm test -- --ci`	73	73	0	0	+60: payment, refresh, errors, operator, paged queue, fixed discount (25 suites)
Total		**1,192**	**1,192**	**0**	**11**	**+354**

Note. Source: log of the *develop* GitHub Actions runs of October 24, 2026 (*gh run view --log*): Maven `Tests run:` lines, *dotnet test* `Passed!`, the Vitest summary, and the Jest summary. Without the mobile app, the total is 1,119. The reasons for each difference are inferred from the branches merged that day. The October 23 table (838 tests) remains in the document's history.

3.4 Live Tests of Payment at Booking (October 26, 2026)

On October 26, 2026 the team tested payment at booking live (payment mode at booking, cash with a limit, and expiry of the digital payment; AC-88), first locally, with all the services running, and then in production. Locally, booking-service ran with the payment window reduced to 1 minute and the expiry job every 15 seconds, so as not to wait 30 minutes. The tests were run through the API and on the web screen. Table 5e shows each test and its result.

Table 5e

Local live tests of payment at booking (October 26, 2026)

#	Test	Expected result	Actual result
1	Book as a customer without `paymentMode`	400 `INVALID_PAYMENT_MODE`	400 `INVALID_PAYMENT_MODE`
2	First cash booking (`CASH`)	201	201
3	Second cash booking while the first is active	409 `CASH_BOOKING_LIMIT`	409 `CASH_BOOKING_LIMIT`
4	`DIGITAL` booking without uploading the receipt	Cancelled automatically when the deadline passes, with the reason `PAYMENT_TIMEOUT`	Cancelled automatically after ~70 s with `PAYMENT_TIMEOUT`
5	`DIGITAL` booking paid on time	Stays `CONFIRMED` and the booking gets `paymentReported=true`	Stayed `CONFIRMED`; `paymentReported=true` arrived in ~5 s through `payment.reported`
6	Pay for a booking already cancelled	409 `BOOKING_NOT_PAYABLE`	409 `BOOKING_NOT_PAYABLE`
7	Choose `PAYMENT_TIMEOUT` when cancelling by hand	The reason cannot be chosen	It cannot be chosen
8	Cancel the cash booking and book in cash again	201: the cash slot is freed	201
9	**Simulated failure 1:** payment-service without RabbitMQ (the `payment.reported` event is lost) and the paid `DIGITAL` booking reaches its deadline	Not cancelled	Not cancelled: the job asked payment-service, which answered `reported=true`, and marked the booking as reported
10	**Simulated failure 2:** payment-service down when an unpaid `DIGITAL` booking expires	Not cancelled while payment does not answer; cancelled when it comes back	Not cancelled ("not cancelled this round: PAYMENT_SERVICE_UNAVAILABLE"); when payment-service came back, the job cancelled it
11	Web screen: step 4 "How will you pay?", cash modal, error of the second cash booking, redirect to pay with the deadline and without the cash option, and "Pay before HH:mm" badge in the history	Each element appears as in the User Manual	Passed

Note. Prepared by the authors from the live test of October 26, 2026 against *develop* of booking-service, payment-service, and the web. The loss of the event was detected in this same test: before the query to payment-service, the `payment.reported` event was lost and the job cancelled a booking that had in fact been paid. It was fixed with the synchronous query before cancelling (AC-89), and failures 1 and 2 were tested with that fix. **Environment finding:** on Windows, .NET resolved `localhost` over IPv6 and Docker Desktop cut payment-service's AMQP connection; with `RABBITMQ_HOST=127.0.0.1` it works. It does not apply in production, because the host is `rabbitmq`.

In production (Azure), on October 26, 2026 around 12:02 (Colombia time), booking-service, payment-service, and the web were deployed from *develop* (Deployment Document, section 1.3.11). All services were healthy and there were no errors in the logs. It was verified that the new columns of `booking.booking` are `NVARCHAR`, that the `PAYMENT_TIMEOUT` reason exists, that the `booking.payment-events` queue has a consumer, that payment's internal path answers 401 without a key and 200 `reported:true` with a key for a booking with an approved payment, and that there were no publishing failures. The team made a cash booking (#12) and a digital payment booking (#13, with its deadline at 13:06).

Table 5f

Automated tests that changed with payment at booking (develop of October 26 and main of October 27, 2026)

Artifact	Tests	Before the change	What they add
booking-service	137 (133 on October 26)	103 (95 in the CI of October 24)	`PaymentPolicyTest` (limit and deadline), `PaymentEventsListenerTest`, `PaymentDeadlineJobTest`, `PaymentServiceDirectoryTest` (answer, error, and internal key), new rules in `BookingServiceTest`, `BookingTest`, and `CancellationReasonTest`, and `BookingApiWebTest`
payment-service	275 with *v1.1.3*: domain 142, application 61, API 47, and integration 25, 12 of them against SQL Server (264 with *v1.1.2* and 247 on October 26, without the integration tests)	220: domain 134, application 52, API 34	`PaymentReported` event (also when reporting again), `reported` query (IN_REVIEW or APPROVED), `InternalApiKeyTests` (401 without a key or with a wrong one, a JWT does not work), and the test of reporting a cancelled booking (`BOOKING_NOT_PAYABLE`). The integration tests run in the CI against SQL Server 2022 since October 27, 2026. *v1.1.3* adds `PaymentConnectionStringTests` (API: without `DB_PASSWORD` the service does not start) and `RabbitMqOptionsTests` (integration: *guest* only against a local RabbitMQ)
Web (Angular)	242 in 68 files (239 on October 26)	212	Payment method step, deadline badge and cash label in the history, payment screen according to the mode, error codes, and Terms and Conditions
Mobile app	106 (100 on October 26)	73	Payment method step (`useReservePaymentStep`), badge, cash label, payment screen according to the mode, and texts (Jest)

Note. Prepared by the authors from the *develop* suites of each repository on October 26, 2026. With the counts of Table 5d for the other artifacts, the total is 1,306 without the tests of payment's infrastructure project (on October 24 it was 1,187 with the same criterion). The

> mobile app got the change with the *v1.1.1* OTA on October 26, 2026. On October 27, 2026 booking-service went up to 137 tests (alternatives that respect the vehicle's other bookings, at most 10 services, and 400 for an invalid business hours week) and customer-service to 83 (owners of many vehicles in one query), for 1,311 with the same criterion, and the end-to-end tests send the payment method. In the `main` CI of October 27, 2026 all of them passed: gateway 62, security-service 200, customer-service 83, booking-service 137, operations-service 110, notification-service 133, and payment-service 275 with *v1.1.3* (1,000 in the backend), web 242, and mobile app 106, for a total of **1,348**, which already includes payment's 25 integration tests (12 against SQL Server). With payment *v1.1.2* the total was 1,337.

4. Functional Test Cases

Cases CP-01 to CP-33 were defined on SRS v4 and first run on October 9, 2026; cases CP-34 to CP-42 cover the requirements implemented afterwards (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037, and RF-038) and the administrator accounts of RF-028. All 42 were run on October 19, 2026, against the full system through the API Gateway (the same calls the web and mobile clients make), with a script that creates its own accounts and data and deletes them at the end. All 42 passed (Figure 7). The changes of October 21, 2026 (the token of a closed session, the close of expired bookings, and the receipt of at most 2 MB) do not have a functional case of their own yet; the automated tests in Table 5 cover them (gap 7 in Table 8). Cases CP-43 to CP-46 were added on October 27, 2026 for payment at booking and the new booking-service validations; the October 19 script did not run them: their result is that of the booking-service automated tests that cover them, green in the `main` CI of October 27 (137 tests).

Table 6

Functional Test Cases

ID	RF	Case	Precondition	Steps	Data	Expected result	
CP-01	RF-001	Successful customer sign-up	Email not registered	Sign up and log in	ID, name, phone, new email, and valid password	Account created with the CLIENT role; the profile is created in customer-service	Sign-up 201; sign-in 200 ` customer-service
CP-02	RF-001	Sign-up with a duplicate email	The email already exists	Sign up with the same email	Email from CP-01	"Email already registered" message; no account is created	409 `EMAIL_ALREADY_RE(` email
CP-03	RF-002	Login by role	CLIENT, OPERATOR, and ADMIN users	Log in with each one	Valid credentials	Redirects to `/client`, `/operator`, or `/admin` by role	The three sign-ins return 2 roles; the web redirects by
CP-04	RF-002	Wrong credentials	—	Log in with a wrong password, then with a nonexistent email	Two attempts	Same generic message in both; it does not reveal whether the email exists	Both: 401 `INVALID_CRED`
CP-05	RF-003	Recover password	Registered user; Mailpit running	Request a code, read it in Mailpit, verify it, and set a new password	6-digit code	Password changed; login with the new one works	6-digit code read in Mailp new password 200
CP-06	RF-003	Code locked after attempts	Code requested	Enter 5 wrong codes, then the right one	5 invalid codes	The right code is no longer accepted	The 5 wrong codes: 400; t
CP-07	RF-004	Password change with a wrong current password	Logged in	Change the password with the wrong current one	Wrong current password	"Incorrect current password" error; nothing changes	400 `INCORRECT_CURRENT`
CP-08	RF-005	Create a booking	Customer with an SUV	Choose vehicle, Basic service, a free date and time and, since October 26, 2026, the payment method (`paymentMode`)	SUV, Basic, `paymentMode` `DIGITAL` or `CASH`	Booking created; price COP 23,000 (seed price of Basic for SUV)	Booking created (201) wi SUV
CP-09	RF-006	Time slot taken	Active bays of a slot booked	Book the same slot until it is full	Several customer vehicles	The booking is not saved; up to five alternative times on the same day are offered	Three bookings took the t `SLOT_UNAVAILABLE` (`NO_`

ID	RF	Case	Precondition	Steps	Data	Expected result	
CP-10	RF-007	Cancel a booking before it starts	Booking CONFIRMED	Cancel as the customer (the customer writes no reason)	—	Booking is CANCELLED with the reason `CUSTOMER_REQUEST` set by the system	200, booking CANCELLI
CP-11	RF-007	Change a booking in progress	Booking IN_PROGRESS	Try to cancel	—	Action blocked with a message	409 `BOOKING_NOT_CHANG`
CP-12	RF-009	Assign an unavailable operator	Operator with an absence at the booking time	Check candidates and try to assign them; then assign an available one	Absent operator	The operator is not allowed; available operators are listed first	The operator without a sh assigning them returns 40 available one: 200
CP-13	RF-011	Operator sees their services	Booking assigned to the operator	Check assigned services	—	List with customer and vehicle data	200; the service carries th
CP-14	RF-012b	Start and finish a service	Assigned service PENDING, 15 min before its time	Another operator tries to start; the assigned one starts and finishes	—	States IN_PROGRESS and COMPLETED; the other operator gets an error	Another operator: 404 `BO` and finishes (200); the bo
CP-15	RF-012	Rate a service once	Booking COMPLETED	Rate 5 stars; try to rate again	5, comment	First rating saved; the second is rejected	First rating 201; the secon
CP-16	RF-013	Payment methods	Payment accounts configured	Check methods (admin) and accounts (customer)	—	Cash, Nequi, Daviplata, and bank transfer; cash shows no QR	Methods: Cash, Nequi, Da accounts; cash without QI
CP-17	RF-013b	Upload a receipt	Booking awaiting payment	Pay with Nequi attaching image, amount, and reference	PNG < 10 MB	Payment created in review with its receipt	201; payment IN_REVIE server
CP-18	RF-014	Approve and reject payments	Payments in review	Approve one and reject another with a reason	Rejection reason	States APPROVED and REJECTED; the customer gets a notification	APPROVED and REJEC and PAYMENT_REJECT
CP-19	RF-016	Duplicate reference	Approved payment with reference R1	Upload another receipt with R1 on another booking	Reference R1	Receipt rejected	409 `TRANSACTION_REFE`
CP-20	RF-018	Booking notification	`MESSAGING_ENABLED=true`	Confirm a booking; open notifications; mark as read	—	The notification arrives; the unread counter goes down	BOOKING_CONFIRME it was marked
CP-21	RF-020	Language saved on the account	Logged in	Switch to English; start a new session	`en`	The UI opens in English	Saved 200; in a new sessi
CP-22	RF-021	Schedule exception	—	Create a "closed" exception and check availability that day	10/20/2026	No times available that day	Exception created (201); t
CP-23	RF-024	Create and deactivate a promotion	Admin session	Create a 10% promotion and deactivate it	10% discount	Promotion visible, then inactive	Created (201) and deactiv
CP-24	RF-026	Earn points	Service with 10 points configured	Approve the booking payment; check the balance	—	The balance grows by the service points	The balance goes up 30 p

ID	RF	Case	Precondition	Steps	Data	Expected result	
CP-25	RF-027	Deactivate one's own account	Customer session	Deactivate the account with the current password and try to log in	Current password	The session is revoked and login is rejected	Deactivation 204; the new
CP-26	RF-028	Create an operator and restrict by role	Admin and customer sessions	Admin creates an operator; the customer calls `GET /admin/users`	Operator data	OPERATOR account created; the customer gets 403	OPERATOR account crea
CP-27	RF-029	Edit the permission matrix	Admin session	Remove `EDIT_DATA` from OPERATOR; try to create a role	Permission `EDIT_DATA`	The matrix is saved; roles cannot be created	Matrix saved (200); creati
CP-28	RF-030	Register a vehicle with a duplicate plate	Customer with a registered vehicle	Register another vehicle with the same plate	Plate FPW653	Registration is rejected	409 `PLATE_ALREADY_RE(`
CP-29	RF-031	Price change without affecting bookings	Confirmed booking with Basic	Admin raises the Basic/SUV price; check the existing booking and create a new one	COP 23,000 → 24,000	The existing one keeps its price; the new one uses the new price	The existing booking keep
CP-30	RF-032	Bay under maintenance	BAY-04 under maintenance (seed)	Fill a slot and check which bays were used	—	The bay under maintenance is not used for new bookings	With Bay 1 under mainter 2 and 3 and the third got 4
CP-31	RF-033	Disallowed status transition	Booking CANCELLED	Admin tries to change it to CONFIRMED	—	Rejected	409 `INVALID_STATUS_TI`
CP-32	RF-034	Deactivated payment account	Active Daviplata account	Admin deactivates it; the customer checks accounts	—	The account is no longer shown to the customer	Account deactivated (200
CP-33	RF-035	Refund an approved payment	Payment APPROVED	Admin refunds	—	Status REFUNDED; `payment.refunded` is published and the customer is notified	Status REFUNDED; the c booking points were rever
CP-34	RF-004b	Lockout after failed attempts	Active account	Sign in 5 times with a wrong password and then with the right one	Email of a new account	At the fifth failure the account is locked for 15 minutes; the right password does not work either	Attempts: 401, 401, 401, 4 `ACCOUNT_LOCKED`
CP-35	RF-008	Automatic operator assignment	Operators with a shift that day	Book without choosing an operator	Basic service, SUV	The booking is assigned to a free operator without the administrator	The booking was assigned
CP-36	RF-010	The customer chooses the operator	The operator has a shift and is free	List the free operators and book choosing one	`preferredOperatorId`	The booking stays with the chosen operator	The chosen operator show
CP-37	RF-015	Payment history with filters	The customer has two payments	Query the history with the day of the paid bookings, with another day, and with an inverted range	Dates	Only that day's payments come out; the inverted range is rejected	With the day of the paid b day, none; the inverted ran

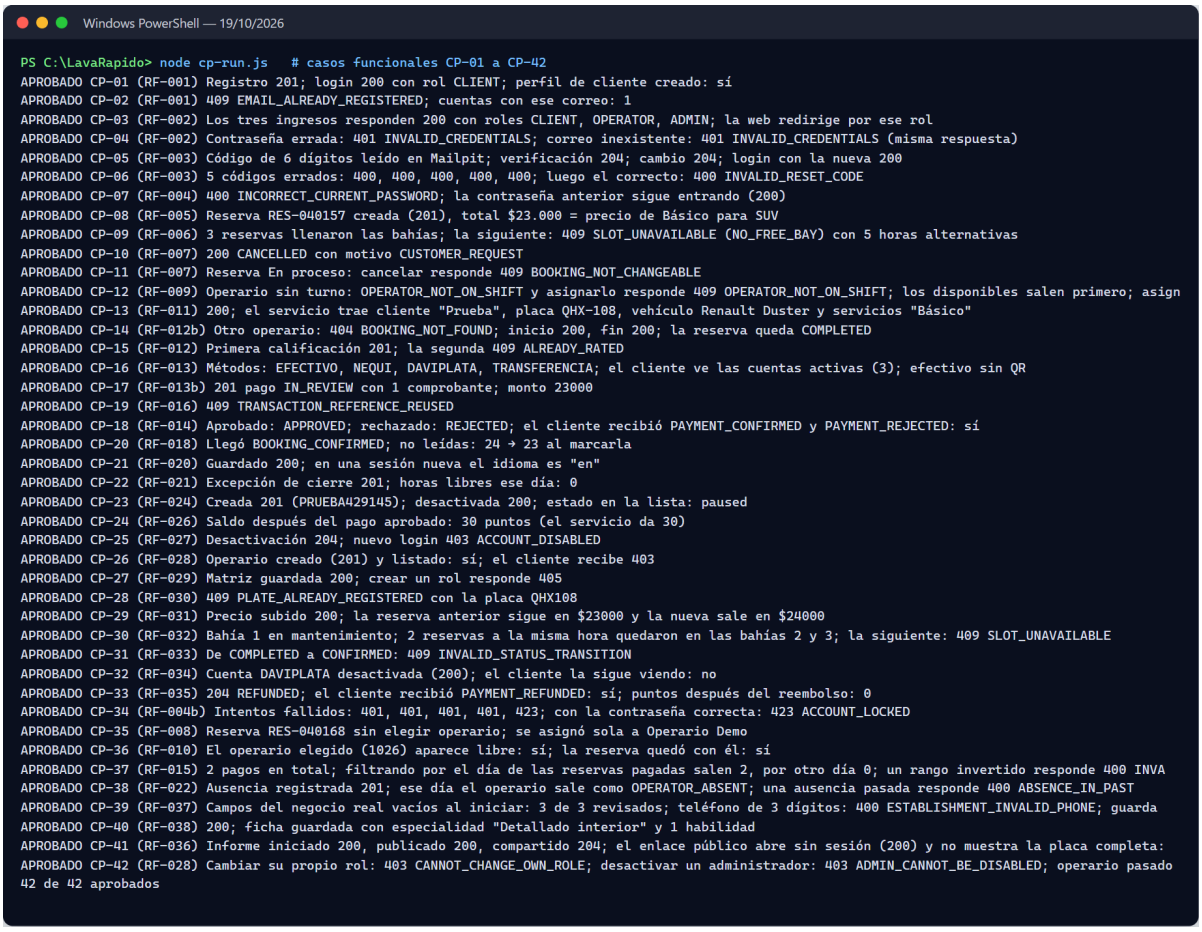
ID	RF	Case	Precondition	Steps	Data	Expected result	
CP-38	RF-022	Operator absence	Operator with a shift	Record an absence that day, list the candidates for a booking that day, and record a past absence	Whole-day absence	The operator shows as absent; the past absence is rejected	Absence recorded (201); t the past absence returns 4
CP-39	RF-037	Business data to be completed	System with migration 111 applied	Read the data, save an invalid phone, and then a valid support line	3-digit phone	The real business data is empty; the invalid phone is rejected; the valid change shows on the public page	The tax data and channels `ESTABLISHMENT_INVALI` public page
CP-40	RF-038	Operator profile	Operator created	Save a specialty and a skill	EXPERT skill	The profile is saved	200; profile saved with its
CP-41	RF-036	Inspection report	Confirmed booking of a 200-minute service	Start the report, finish a phase, publish, share, and open the public link	Temporary 200-minute service	The customer gets the notice and the link opens without a session with the plate masked	Report started (200), publ INSPECTION_REPORT the full plate
CP-42	RF-028	An administrator account cannot be deactivated	Administrator session	Try to deactivate an administrator account	—	Rejected: an administrator account cannot be deactivated	403 `ADMIN_CANNOT_BE_[`
CP-43	RF-005	Payment mode is required	Customer with a vehicle	Book without `paymentMode` and then with a mode that does not exist	No `paymentMode`; `CARD`	The booking is rejected with 400 `INVALID_PAYMENT_MODE` and is not saved	Meets (automated test gre `BookingServiceTest`, a `BookingApiWebTest.pa`
CP-44	RF-005	Cash booking limit	Customer with one active cash booking made by themselves	Make another cash booking	`paymentMode CASH`	409 `CASH_BOOKING_LIMIT`; nothing is saved	Meets (automated test gre `BookingServiceTest`, a `BookingApiWebTest.pa`
CP-45	RF-005	At most 10 services per booking	Customer with a vehicle; administrator session	Book with `serviceIds [null]` and with 11 services; reschedule as administrator with `[null]`	`serviceIds [null]` and `[1..11]`	400 `VALIDATION_ERROR` (500 before); the use case is not called	Meets (automated test gre `BookingApiWebTest.nu`
CP-46	RF-021	Invalid day in the weekly opening hours	Administrator session	Save the opening-hours week with a day that does not exist and then with an empty list	`dayOfWeek` 9; `[]`	400 `VALIDATION_ERROR` (500 before); the opening hours do not change	Meets (automated test gre `BookingApiWebTest.an`

Note. On October 19, CP-42 also tested the role change of an account (403 `CANNOT_CHANGE_OWN_ROLE` when changing one's own role, and an operator turned into a customer with 200). That part was removed from the case on October 20, 2026, when the feature was reverted in the backend, the web and the app.

Note. The prices in the test data come from the seed changelogs (`103-seed-catalog.sql`, `104-seed-booking-catalogs.sql`). The script creates a test customer, two operators, and an administrator, with emails of the `prueba.lavarapido.test` domain, and its own vehicles and bookings. At the end it deletes everything it created and restores the Basic price, the permission matrix, the bay, the payment account, and the business data; the database was checked to have nothing left. The recovery emails were read in Mailpit. Evidence: execution log with the HTTP code and the error code of each step (Figure 7).

Figure 7

Run of the 42 Functional Cases (October 19, 2026)



5. Traceability Matrix

Of the 41 requirements in SRS v8.4 (3 of them partial as of October 27, 2026: RF-005, RF-015, and RF-023, with the reason in the SRS; on October 23 there were 11), 38 have a functional case and 39 a related automated test; RF-032 and RF-034 only have a functional case, and RF-017, RF-023, and RF-025 only an automated test.

Table 7

RF → Status → Endpoint / Screen → Test Case

RF	Status (SRS v8.4)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related
RF-001	Implemented	`POST /auth/register`	`/auth/register`	CP-01, CP-02	`AccountRegistratic` `UserRegisteredList`
RF-002	Implemented	`POST /auth/login`	`/auth`	CP-03, CP-04	`AuthenticationServ` `JwtTokenIssuerTest`
RF-003	Implemented	`POST /auth/password/forgot`, `/verify`, `/reset`	`/auth/forgot-password`	CP-05, CP-06	`PasswordRecoverySe` `PasswordResetToker`
RF-004	Implemented	`PUT /users/me/password`	`/client/profile`	CP-07	`UserProfileAndAdmi` `PasswordPolicyTest`
RF-004b	Implemented	`POST /auth/login` (423 `ACCOUNT_LOCKED`)	`/auth`	CP-34	`AuthenticationServ` window), `PasswordRe` reset lifts it), `ApiSecur`

RF	Status (SRS v8.4)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related
RF-027	Implemented	`GET/PATCH /users/me`, `GET/PUT /users/me/address`, `PUT /users/me/email`, `POST /auth/logout`, `POST /users/me/deactivate`, `GET /auth/session` (gateway)	`/client/profile`	CP-25	`UserProfileAndAdmi` `PersonAddressServi` `AuthenticationServ` `ApiSecurityWebTest` `SessionCheckFilter` session), `auth.spec.t`
RF-028	Implemented	`GET/POST /admin/users`, `PATCH /admin/users/{id}/status`	`/admin/management`, `/admin/operators`	CP-26, CP-42	`UserProfileAndAdmi` `ApiSecurityWebTest`
RF-029	Implemented	`GET /admin/roles-permissions`, `PUT /admin/roles-permissions/{role}`	`/admin/management`	CP-27	`RolePermissionAdmi` `management.spec.ts`
RF-030	Implemented	`/vehicles`, `GET /vehicle-types`, `GET /admin/vehicles`	`/client/vehicles`	CP-28	`CustomerVehicleSer` `LicensePlateTest`, `C`
RF-031	Implemented	`GET /catalog/services`, `/admin/catalog/services`	`/admin/management`	CP-29	`CatalogServiceTest`
RF-021	Implemented	`PUT /admin/schedule/business-hours`, `POST /admin/schedule/exceptions`	`/admin/schedule`	CP-22, CP-46	`AvailabilityPolicy` `BookingApiWebTest`
RF-032	Implemented	`/admin/bays`	`/admin/schedule`	CP-30	—
RF-037	Implemented	`GET/PUT /admin/establishment`, `GET /establishment`	`/admin/settings`	CP-39	`EstablishmentTest`,
RF-005	Partial	`POST /bookings`, `GET /catalog/services`	`/client/reserve`	CP-08, CP-43 to CP-45; live test of October 26 (Table 5e)	`BookingServiceTest` `PaymentPolicyTest`, `PaymentDeadlineJob` `PaymentEventsListe` `PaymentServiceDire` `form.spec.ts`
RF-006	Implemented	`GET /bookings/availability`	`/client/reserve`	CP-09	`AvailabilityPolicy`
RF-007	Implemented	`PUT /bookings/{id}`, `POST /bookings/{id}/cancel`	`/client/history`	CP-10, CP-11	`BookingTest`, `Bookir`
RF-033	Implemented	`GET /admin/bookings`, `PATCH /admin/bookings/{id}/status`, `GET /admin/assignments/to-close`, `POST /admin/assignments/{bookingId}/attended`	`/admin/reservations`, `/admin` ("To close")	CP-31	`BookingTest`, `Bookir` the cancellation of expi the internal no-show ro `FollowUpPolicyTest` `reservations-store`
RF-008	Implemented	`booking.confirmed` event, `GET /internal/v1/bookings`	`/admin/reservations`	CP-35	`AutoAssignmentPoli` `OperationsServiceT` deactivated account), `b`
RF-009	Implemented	`GET /admin/assignments/{id}/candidates`, `PUT /admin/assignments/{id}`	`/admin/reservations`	CP-12	`AssignmentPolicyTe` `OperationsServiceT` rule for bookings witho `reservations-store`
RF-010	Implemented	`GET /operators/available`, `POST /bookings` (`preferredOperatorId`)	`/client/reserve`	CP-36	`OperationsServiceT` `OperationsApiWebTe` `flow.test.js`
RF-011	Implemented	`GET /operator/services`	`/operator/assigned-services`	CP-13	`OperationsServiceT`
RF-012b	Implemented	`POST /operator/services/{id}/start`, `/finish`	`/operator/assigned-services`	CP-14	`StartPolicyTest`, `Se`
RF-012	Implemented	`POST /ratings/{bookingId}`	`/client/history`	CP-15	`ServiceExecutionTe`
RF-022	Implemented	`PUT /admin/operators/{id}/availability`, `/absences`, `GET /admin/operators/summary`	`/admin/operators/:id`	CP-38	`AssignmentPolicyTe` `TeamSummaryCalcula`
RF-036	Implemented	`/admin/inspections/{bookingId}/**`, `GET /public/inspections/{token}`	`/admin/reservations/:id/inspection`, `/report/:token`	CP-41	`InspectionReportTe` `OperationsApiWebTe`
RF-038	Implemented	`GET /admin/operators/profiles`, `PUT /admin/operators/{id}/profile`	`/admin/operators/:id`	CP-40	`OperatorProfileSer`
RF-034	Implemented	`/admin/payment-accounts`, `GET /payment-accounts`	`/admin/settings`	CP-32	—

RF	Status (SRS v8.4)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related a
RF-013	Implemented	`GET /payment-accounts`	`/client/payment`	CP-16	`PaymentQueryServic`
RF-013b	Implemented	`POST /payments`	`/client/payment`	CP-17	`PaymentTests` (receip `PaymentCommandServ`
RF-014	Implemented	`POST /admin/payments/{id}/approve`, `/reject`	`/admin/payments`	CP-18	`PaymentTests`, `Payme`
RF-015	Partial	`GET /payments/me?from=&to=&service=`	None: `/client/history` is the booking history	CP-37	`PaymentQueryServic`
RF-016	Implemented	`POST /payments`	`/client/payment`	CP-19	`PaymentPolicyTests` `PaymentCommandServ` reference)
RF-035	Implemented	`POST /admin/payments/{id}/refund`	`/admin/payments`	CP-33	`PaymentTests`, `Payme`
RF-017	Implemented	`GET /admin/bookings/day-summary`, `GET /operators/my-bookings` and dashboard stores	`/client`, `/operator`, `/admin`	—	`BookingServiceTest` smoke tests (`dashboar`
RF-018	Implemented	`/notifications/*` + RabbitMQ events	`/*/notifications`	CP-20	`NotificationUseCas` `NotificationFactor` `BookingReminderSer` `LocalizedNotificat` `UnassignedBooking` `StaffNotifications`
RF-019	Implemented	`PUT /users/me/preferences`	`/*/configuration`	Covered by CP-21	`UserPreferenceServ`
RF-020	Implemented	`PUT /users/me/preferences`	`/*/configuration`	CP-21	`UserPreferenceServ` `UserPreferenceTest`
RF-023	Partial	`GET /admin/operators/{id}/metrics`, `GET /admin/operators/summary`, payment and booking stores	`/admin/reports`, `/admin/operators/:id`	—	`OperatorMetricsCal` `TeamSummaryCalcula` `reports.spec.ts` (or count)
RF-024	Implemented	`/admin/promotions`	`/admin/management`	CP-23	`PromotionTests`
RF-025	Implemented	`POST /loyalty/redeem`	`/client/payment`	—	`CustomerPromotion`
RF-026	Implemented	`GET /loyalty/balance`	`/client`	CP-24	`LoyaltyTransactior`

Note. The automated test column points to the file whose name and methods match the requirement; code coverage was not measured. `booking-flow.test.js` is in `lavarapido-infra/e2e` and needs the system running.

6. Findings and Open Items

Table 8

Testing Gaps Found

#	Gap	Recommendation
1	The mobile app had no automated tests	Closed: 106 tests with *jest-expo* in the `main` CI of October 27, 2026 (73 in 25 suites on October 24); coverage is not measured yet
2	Many of the web tests only checked that the component is created	As of October 27, 2026 there are 242 tests in 68 files (212 in 66 on October 24), with the real guard and interceptor, the coupon payment, the refresh, and the translations, and coverage in the CI; the booking form is still missing
3	The 11 payment persistence tests (7 until October 23) need a migrated database and skip without it; the first 7 were run on October 19 and 21 against `LavaRapidoPaymentIT` and passed; those for the indexes of migration 023 had not been run	Closed for payment-service on October 27, 2026: they are 25 integration tests with *v1.1.3* (12 against the database) and they run in the CI, in the "Pruebas de integración" job, against SQL Server 2022 with the Liquibase migrations, without skipping any. Doing the same in the other services is still pending
4	For RF-023 (reports) the operator metrics, the team summary and, since October 21, 2026, that only completed bookings count (`reports.spec.ts`) are tested; the PDF/Excel export has no test	Add a functional case for reports and the PDF/Excel export
5	RF-032 (bays) and RF-034 (payment accounts) have no dedicated automated tests: `ScheduleService` and `PaymentAccountService` have no test class	Add unit tests for those two services
6	The requirements implemented after October 9 (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037, and RF-038) had no functional case of their own	Closed on October 19, 2026, with cases CP-34 to CP-41
7	The changes of October 21, 2026 (the token of a closed session stops working, bookings that ended and nobody closed with the "To close" list, the receipt over 2 MB, and the cash option that only appears with its account active) have no functional case of their own; automated tests cover them, except the cash option, the 19 end-to-end tests were run with the session check on, and "To close" was checked live on the web and in the mobile app	Add those functional cases and run them against the complete system; add a `PaymentAccountService` test for the cash account
8	The changes of October 24, 2026 (server amount and `AMOUNT_CHANGED`, `BOOKING_SERVICES_LOCKED`, `VEHICLE_HAS_ACTIVE_BOOKINGS`, consent, `/auth` limit, `TOO_MANY_ATTEMPTS`, 4-hour window to start, operator notes) have no functional case of their own; the automated tests and the remote battery of `lavarapido-infra/e2e/remote` (44 cases) cover them, but the battery has not been run against the public server yet	Create the three test accounts and the `E2E_*` secrets, run *e2e.yml* without flags and then with `allow_writes`, and add the corresponding functional cases
9	Migrations 044, 045, and 113 (booking), 047, 116, and 048 (notification), and 022 and 023 (payment) were only tested when applied on the public server on October 24, 2026. 047 was applied without error, but notification could not read `channel` or `language` (VARCHAR read as NCHAR because of `use_nationalized_character_data`) and stopped sending push notifications and emails; the unit tests, which run without SQL Server, did not catch it, and the live test found it. 048 (NVARCHAR) was first tested in `tempdb` with the data kept and deployed that day at 17:12 UTC (AC-87)	Integration tests against SQL Server in the CI (RNF-003): the 047 failure shows they are needed (since October 27, 2026 payment-service's run; notification's and the other services' are missing); meanwhile, a live test after each deployment. Booking migration 049 and seed 117 (October 26, 2026) were tested this way, locally and in production: columns in `NVARCHAR` and the `PAYMENT_TIMEOUT` reason present
10	The loss of an event between services was only detected in the live test of October 26, 2026 (the paid booking that was cancelled, Table 9, defect 13): there is no Outbox nor an automated test that simulates it between services	Keep the query to payment-service before cancelling (AC-89), covered by `PaymentDeadlineJobTest` and `PaymentServiceDirectoryTest`; add payment at booking to the end-to-end battery, and the Outbox (AP-02)

Table 9

Defects Found While Running the System

#	Defect	Evidence	Severity
1	The administrator dashboard always says "¡Hola, Laura!" regardless of who logs in	`Web/src/app/features/admin/pages/dashboard/dashboard.ts:73` (`adminName = 'Laura'`); "Laura Méndez" is also hard-coded in `payment-review-modal.html:167`	Low
2	In the operator's "Assigned services" (web), the Customer column shows the license plate	`Web/src/app/features/operator/pages/assigned-services/assigned-services.ts:141`	Medium
3	On the customer home (mobile), "Assigned operator" is empty even though the booking has an operator	Booking RES-060003, assigned in CP-12; mobile screenshot	Medium
4	The brand name is inconsistent: "AquaWash" (mobile), "Car Wash" and "Lavado Vehicular" (web), "Express Car Wash" (establishment)	Web and mobile screenshots	Low
5	The landing page advertises a "mobile wash service" and "wherever you are", but SRS v4 defines a single-location car wash	Landing screenshot; SRS v4, section 1	Low
6	If there are two accounts of the same method (two Nequi), the payment screen shows both selected at once	Customer Payments screenshot	Low
7	When logging out, the web and the app showed "Your session expired", because a 401 that arrived after the token was deleted was taken as an expired session	`auth-interceptor.ts` (web) and `AuthServiceContext.tsx` (mobile)	Medium
8	On a phone (360 px) there were buttons off the screen in the landing bar, overlapping titles, the ☰ button over the titles, Assigned services badges off the screen, Management filters outside their card, and Settings switches over their text	Responsive design check of October 17, 2026 (Table 5c)	Medium
9	In the mobile app, the date and search inputs were squashed when placed in a column (History and Ratings)	`shared/components/screen/FilterInputs.tsx`	Low
10	The customer's next service progress showed 10% or 25% although the wash had not started	`features/client/pages/dashboard/dashboard.ts` (web) and `modules/client/models/client.ts` (mobile)	Low
11	The first version of the automatic close (October 21) set to No-show every confirmed booking nobody started one hour after its end; if the operator washed the car and forgot to press "Start", the history, reports, and metrics recorded that the customer did not come, and No-show is final	Team review of October 21, 2026; `ExpiredBookingsJob` and `Booking.expire` in booking-service	High
12	A `null` field in an event payload sent the whole message to notification-service's dead-letter queue (DLQ), and the notice was not created	`DomainEventEnvelope.java` in notification-service	Medium
13	When the `payment.reported` event was lost, the expiry job cancelled with `PAYMENT_TIMEOUT` a booking with digital payment that had in fact been paid	Live test of October 26, 2026 (Table 5e); `PaymentDeadlineJob` in booking-service	High

Note. Defects 7 to 10 were fixed between October 16 and 17, 2026, on the *develop* branch of the web and the app; 8 and 9 were measured again with no overflow (Table 5c). Defects 11 and 12 were fixed on the evening of October 21, 2026: the system now only cancels a scheduled booking that expired without confirmation and sets to No-show a confirmed one that never had an operator; one that had an operator goes to "To close" so the administrator decides (11, `FollowUpPolicyTest`, `FollowUpServiceTest`); and the `null` field is ignored (12, `DomainEventEnvelopeTest`). Defect 13 was fixed on October 26, 2026, before deploying: the job asks payment-service before cancelling and, if payment does not answer, it does not cancel in that round (AC-89, `PaymentDeadlineJobTest`, `PaymentServiceDirectoryTest`; simulated failures 1 and 2 of Table 5e).

Note. Code review of October 16, 2026: the six defects are fixed. The dashboard greets with the session name and the payment review shows the reviewer (1); the Client column shows the customer name (2); the customer home shows the assigned operator on the web and on mobile (3); the brand is LavaRápido on the web and on mobile (4); the home page describes the service at the car wash (5); and the payment screen selects by account and shows the holder when two accounts share a method (6, `payment.spec.ts`).

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